

PlateLoop: expansion plan

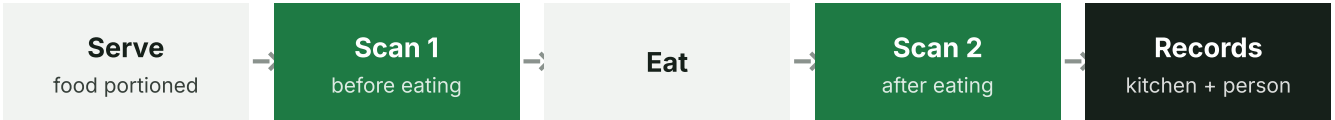
How a canteen tray scanner grows from one school to hospitals, offices and individual users.

PlateLoop is a tray scanner for canteens. It records each tray before and after a meal, identifies the dishes and works out how much of each was eaten and the nutrients it contained. Kitchens use the records to plan how much food to prepare; the people eating receive their own intake and nutrition feedback through an app designed for their setting.

This report sets out the order in which PlateLoop would expand, the setting and users at each stage, and the published evidence for each. It contains no financial projections.

790,000 t food waste generated in Singapore, 2025 [1]	18% of food waste recycled, 2025 [1]	-36% food waste after catering sites began measuring it, year one [2]	317 MOE primary and secondary schools [10]
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1 What PlateLoop does



Scan 1 – Scan 2 = what was eaten, per dish (e.g. 620 g served – 95 g left = 525 g eaten)

Figure 1. Each tray is scanned twice. The difference between the scans is what was eaten.

EACH SCAN RECORDS	HOW	USED BY
Food type	Overhead colour and depth camera, matched against the day's menu	Kitchen: which dishes are eaten or left
Amount	Food volume from the depth camera, checked by a weighing platform	Kitchen: how much to prepare and order
Nutrients	Grams of each dish matched to the kitchen's recipes	Person: energy, protein, carbohydrate, fat, fibre, sodium

2 The problem

Singapore generated 790,000 tonnes of food waste in 2025, more than in 2023, and the share recycled has stayed at 18–19% since 2021 [1]. Canteens plan food by headcount and experience because they have no record of what each diner eats.

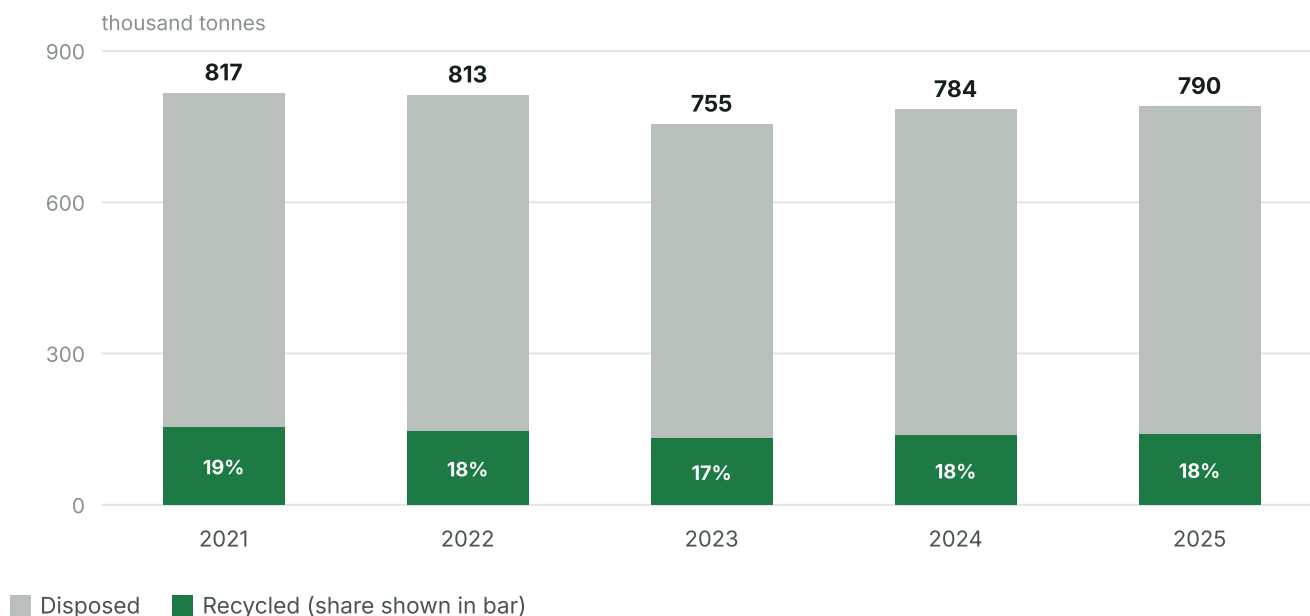


Figure 2. Food waste generated in Singapore, 2021–2025, thousand tonnes. Source: NEA [1].

3 Evidence for the approach

Measuring food waste reduces it. Across 86 catering sites in six countries, including Singapore, sites that measured their food waste reduced it by 36% by weight within a year; 64% recovered their investment in the first year and 79% spent less than US\$10,000 [2].



Index, average across 86 catering sites in six countries including Singapore [2]

Figure 3. Food waste before and after one year of measurement, indexed to 100. Source: Champions 12.3 [2].

Recognising food from tray images is also established. A system using overhead depth images of plated meals identified foods with 88.9% accuracy, and nutrient estimates from food volume closely matched estimates from weight (r^2 0.92–0.99) [3].

4 Expansion plan

PlateLoop expands in six stages. Each stage keeps the same scanner and kitchen dashboard and adds the app for the people eating in that setting.

SETTING	USERS	APP	REASON FOR THIS STAGE
1 One school canteen	Canteen operator; students	Kitchen dashboard; Loopi	Measure real waste and test daily use at one site
2 The caterer's other canteens	Contract caterer	Kitchen dashboard (multi-site)	One agreement covers many canteens; caterers must now separate food waste [11]
3 More schools; university campuses	Students; campus food services	Loopi	317 MOE primary and secondary schools [10]; 13.6% of university students meet fruit and vegetable guidance [8]
4 Hospitals	Patients; nurses; dietitians	Loopi Care	29% of inpatients malnourished [4]; 93% of food charts incomplete [5]; 30% median plate waste [6]
5 Offices	Office workers; canteen operator	Loopi Work	6 in 10 residents eat out 4+ times a week [7]; obesity 12.7% [9]
6 Everyone	Individuals	Phone camera app	Before and after photos of any meal, outside canteens

	Stage 1 One school canteen	Stage 2 The caterer's canteens	Stage 3 More schools and campuses	Stage 4 Hospitals	Stage 5 Offices	Stage 6 Personal app
Tray scanner						
Kitchen dashboard						
Multi-site view						
Loopi (students)						
Loopi Care (patients, nurses)						
Loopi Work (office workers)						
Phone camera app						

Figure 4. Components used at each stage. The scanner and kitchen dashboard are shared by stages 1–5; each sector adds one app.

5 Risks

RISK	RESPONSE
Mixed dishes are harder to recognise	Known daily menu, compartment trays, weighing platform check.
Trays not scanned after eating	Scanner at the tray return; dashboard flags trays still out.
Queues slow down	About 2 s per scan; second scanner for large canteens.
Privacy concerns	No photos leave the scanner; consent first; face sign-in optional.
Results differ from published studies	Stage 1 measures before and after at one site first.

6 Indicators for each setting

The figures below are the published indicators behind stages 3–5. Each describes a problem that per-tray records address: unmeasured intake in hospitals, high plate waste, and diets that diners cannot see.

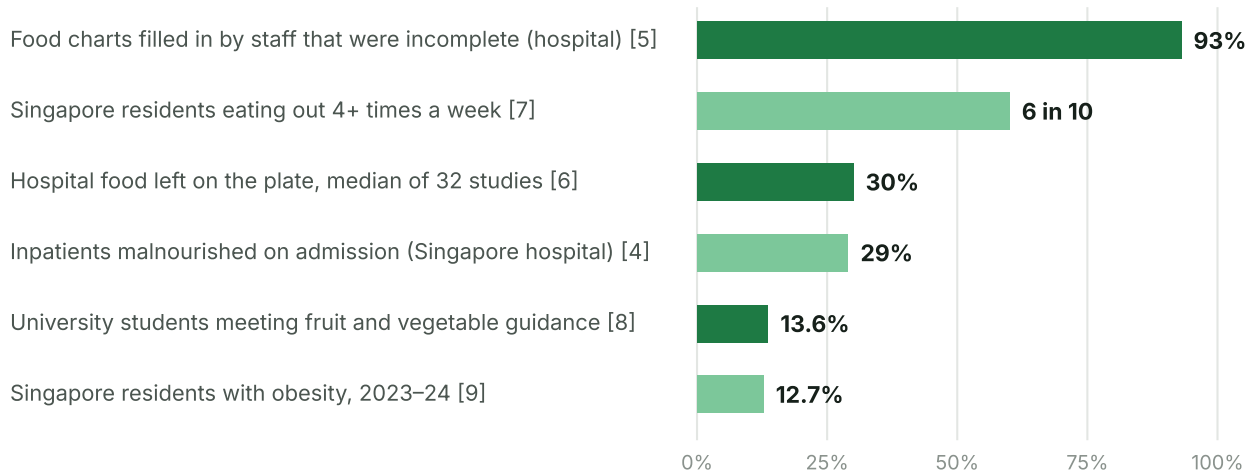


Figure 5. Selected indicators by setting. Reference numbers in brackets.

7 References

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